

6	(a)	(i)		current = $\frac{\text{power}}{\text{p.d.}} = \frac{6.0}{24} = 0.25 \text{ A}$	A1
		(ii)		p.d. = $\frac{24}{12} = 2.0 \text{ V}$	A1
		(iii)		resistance = $\frac{\text{p.d.}}{\text{current}} = \frac{2.0}{0.25} = 8.0 \Omega$	B1
	(b)	(i)		Break occurs in the connection between P, switch, battery and Q.	B1
		(ii)		Break occurs in light bulb H / filament of H has melted/broken	B1
		(iii)		Break occurs somewhere along the wires connecting the bulbs	B1
	(c)	(i)		p.d. = $\frac{24}{11} = 2.2 \text{ V}$	A1
		(ii)		new power = $\frac{V^2}{R} = \frac{2.182}{8} = 0.5951 \text{ W}$ original power = 0.50 W fractional increase = $\frac{0.5951 - 0.50}{0.50} = 0.19$	C1 A1
[Total: 9]					